

III B. Tech I Semester Regular Examinations, Dec/Jan – 2022-23**DEVOPS**

CSE(AIML), CSE(AI), CSE(DS), CSE(AIDS), AIDS, AIML

Time: 3 hours

Max. Marks: 70

Answer any **FIVE** Questions **ONE** Question from **Each unit**

All Questions Carry Equal Marks

UNIT-I

1. a) What is SDLC? Explain various phases involved in SDLC. [7M]
 b) “Agile methodologies apply four values in different ways to guide the development and delivery of high-quality working software” – Support this statement with an explanation of these four values. [7M]
 (OR)
- 2 a) “There are principles of agile methodology that describe the culture in which change is welcome and the customer is the focus of work” – Support this statement with a short description on the principles of agile methodology. [10M]
 b) What advantages does SDLC offer an organization? Explain. [4 M]

UNIT-II

3. a) Describe various components that constitute the DevOps architecture with a neat diagram. [7M]
 b) Write a short note on various categories of waste and hardship in implementing Lean S/W development. [7M]
 (OR)
4. a) What is a DevOps pipeline? Explain various stages involved in the DevOps pipeline. [7M]
 b) Explain the three ways of underpinning principles from which the DevOps behaviors and patterns are derived. [7M]

UNIT-III

5. a) Write a short note on the following terminology related to DevOps [10M]
 - i. Infrastructure as a code
 - ii. Containerization
 - iii. Microservices
 - iv. Cloud Infrastructure
 - v. Orchestration
- b) What is the difference between Agile and DevOps? [4M]

(OR)

6. a) “DevOps implementation strategy is based on six fundamental factors to operate and manage applications” – Support this statement with a proper explanation. [7M]
 b) Explain various factors that influence DevOps Adoption. [7M]

UNIT-IV

7. a) What is Continuous Integration? Explain a few benefits the software industry gets by incorporating Continuous Integration. [7M]
 b) Explain a minimum of seven metrics that must be monitored before implementing Continuous Delivery into our organization. [7M]

(OR)

1 of 2

8. a) “The continuous integration process is a critical component of DevOps which involves a lot of challenges” – Support this statement with an explanation of various challenges and best practices that can help us to succeed. [7M]
- b) Differentiate between [7M]
- i. Continuous Integration and Continuous Delivery
 - ii. Continuous Integration and Continuous Deployment
 - iii. Continuous Delivery and Continuous Deployment
- UNIT-V**
9. a) “Different assessment providers are used at different stages for evaluating the DevOps maturity of an organization” – Support this statement with an explanation of various assessment metrics. [7M]
- b) What is DevOps maturity model? Explain various phases of DevOps maturity model with a neat sketch. [7M]
- (OR)
10. a) Describe the four key tenets of DevOps that all mature ecosystems need to be considered as a benchmark for assessing the maturity of DevOps practices. [10M]
- b) Write a short note on the business benefits of DevOps maturity. [4M]

Code No: 155EX

R18

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

B. Tech III Year I Semester Examinations, January/February - 2023

DEVOPS

(Computer Science and Engineering – Data Science)

Time: 3 Hours

Max. Marks: 75

Note: i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

- 1.a) How is Devops different from agile methodology? [2]
- b) Explain about the artifact repository. [3]
- c) What is the principle of cohesion? [2]
- d) Explain in detail about coupling. [3]
- e) What is Gerrit? [2]
- f) Discuss about shared authentication. [3]
- g) What is Jenkins file system layout? [2]
- h) Explain about Jenkins plugins. [3]
- i) What is unit testing? [2]
- j) Why are there so many deployment systems? Explain. [3]

PART – B

(50 Marks)

2. Explain about Scrum, Kanban, and the delivery pipeline in detail. [10]
- OR**
3. Discuss in detail about release management and agile development model. [10]
4. Describe Devops, architecture and resilience in detail. [10]
- OR**
5. Write a brief note on software architecture. Explain about the monolithic scenario. [10]
6. Explain about different Git server implementations and Docker intermission. [10]
- OR**
7. Discuss about hosted Git servers and the pull request model. [10]
8. Describe the host server and build slaves in detail. [10]
- OR**
9. What are triggers? Explain about job chaining and build pipelines. [10]
10. Explain the following:
a) Deploying with SaltStack b) Testing backend integration points. [5+5]
- OR**
11. Explain the following:
a) Visualization stacks b) Automated integration testing. [5+5]

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